

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12398165
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Mirica; Katherine A. et al.

Two-dimensional stimuli-responsive covalent organic frameworks with high intrinsic conductivity

Abstract

In an embodiment, the present disclosure pertains to a method of sensing an analyte in a sample by: (1) exposing the sample to an electrode that includes a covalent-organic framework with a plurality of metal-coordinated aromatic units that are linked to one another by aromatic linkers; (2) detecting a change in a property of the electrode; and (3) correlating the change in the property to the presence or absence of the analyte. In an additional embodiment, the present disclosure pertains to said covalent-organic frameworks. Additional embodiments of the present disclosure pertain to methods of making the covalent-organic frameworks.

Inventors: Mirica; Katherine A. (Hanover, NH), Meng; Zheng (West Lebanon, NH), Stolz; Robert M. (Lyme, NH)

Applicant: Trustees of Dartmouth College (Hanover, NH)

Family ID: 1000008781340

Assignee: Trustees of Dartmouth College (Hanover, NH)

Appl. No.: 18/126871

Filed: March 27, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230348513 A1	Nov. 02, 2023

Related U.S. Application Data

continuation parent-doc US 16855819 20200422 US 11634446 child-doc US 18126871
us-provisional-application US 62837118 20190422

Publication Classification

Int. Cl.: G01N15/04 (20060101); C07F15/04 (20060101); G01N27/12 (20060101)

U.S. Cl.:

CPC C07F15/045 (20130101); G01N27/126 (20130101);

Field of Classification Search

CPC: C07F (15/045); C07F (15/025); C07F (15/065); C07F (1/005); G01N (27/126); G01N (27/127)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
11634446	12/2022	Mirica	502/159	C07F 15/065
2012/0152845	12/2011	LeVan	210/660	B01J 20/3265
2014/0011286	12/2013	Potyrailo	427/466	G01N 33/0031
2015/0231622	12/2014	Kitagawa	502/152	B01J 37/0209
2016/0229873	12/2015	Talin	N/A	B01J 31/16
2017/0073364	12/2016	Dinca	N/A	H01G 11/30

Primary Examiner: Fortich; Alvaro E

Attorney, Agent or Firm: Shackelford, McKinley & Norton, LLP

Background/Summary

BACKGROUND

(1) Current chemiresistive sensors face several challenges, including limited conductivity, limited sensitivity, and limited stability. Various embodiments of the present disclosure addresses the aforementioned challenges.

SUMMARY

(2) In an embodiment, the present disclosure pertains to a method of sensing an analyte in a sample. In some embodiments, the method includes exposing the sample to an electrode that includes a covalent-organic framework. In some embodiments, the covalent-organic framework includes a plurality of metal-coordinated aromatic units that are linked to one another by aromatic linkers. In some embodiments, the method further includes detecting a change in a property of the electrode and correlating the change in the property to the presence or absence of the analyte.

(3) In another embodiment, the present disclosure pertains to covalent-organic frameworks that include a plurality of metal-coordinated aromatic units that are linked to one another by aromatic linkers. In some embodiments, the covalent-organic framework is in the form of a fully aromatic and conjugated structure.

(4) Additional embodiments of the present disclosure pertain to methods of making the covalent-organic frameworks of the present disclosure. In some embodiments, the methods of the present

disclosure include conjugation of metal-coordinated aromatic units through the utilization of aromatic linkers. In some embodiments, the conjugation occurs through aromatic annulation.

Description

DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A illustrates a method of sensing an analyte in a sample according to an aspect of the present disclosure.
- (2) FIG. 1B depicts a covalent-organic framework (COF) according to an aspect of the present disclosure.
- (3) FIG. 2 illustrates a synthetic route for two-dimensional (2D) conductive COF material, termed COF-DC-8, with nickelphthalocyanine and pyrene subunits connected by pyrazine rings.
- (4) FIGS. 3A, 3B, 3C, and 3D illustrate the structure and characterization of COF-DC-8. FIG. 3A shows top and side view of the structure of COF-DC-8 with 2×2 square grids in eclipsed stacking mode with a Lieb lattice. FIG. 3B shows a comparison of the simulated and experimental powder X-ray diffraction (PXRD) patterns of COF-DC-8. FIG. 3C shows a scanning electron microscopy (SEM) of COF-DC-8. FIG. 3D shows a transmission electron microscopy (TEM) of COF-DC-8.
- (5) FIGS. 4A and 4B illustrate calculated electronic band structure of COF-DC-8 (FIG. 4A) and the corresponding first Brillouin zone (FIG. 4B).
- (6) FIGS. 5A, 5B, 5C, 5D, 5E, 5F, 5G, and 5H illustrate chemiresistive responses of devices with integrated COF-DC-8 to (FIG. 5A) NH₃ (ammonia) (FIG. 5B) H₂S (hydrogen sulfide), (FIG. 5C) NO (nitric oxide), and (FIG. 5D) NO₂ (nitrogen dioxide) under an applied voltage of 1.0 V and an atmosphere of dry nitrogen. Also shown are responses ($-\Delta G/G_{sub.0}$) of COF-DC-8 after 1.5 min (squares) and 30 min (circles) exposure versus concentration of (FIG. 5E) NH₃, (FIG. 5F) H₂S, (FIG. 5G) NO, and (FIG. 5H) NO₂.
- (7) FIGS. 6A and 6B illustrate the initial rate of response as a function of concentration of (FIG. 6A) reducing gases NH₃ and H₂S and (FIG. 6B) oxidizing gases of NO and NO₂.
- (8) FIGS. 7A, 7B, 7C, and 7D illustrate electron paramagnetic resonance (EPR) spectrum of COF-DC-8 before (dashed lines) and after dosing (solid lines) with (FIG. 7A) NH₃ and H₂S, and (FIG. 7B) NO and NO₂. For comparison, the intensity of the peaks has been normalized based on the peak intensity before analyte dosing. FIG. 7C shows X-ray photoelectron spectroscopy (XPS) of S 2p range of COF-DC-8 after H₂S exposure. FIG. 7D shows XPS of N 1s range of COF-DC-8 after NO₂ exposure.
- (9) FIG. 8 illustrates high resolution mass spectroscopy analysis of 2,3,9,10,16,17,23,24-octatosylaminophthalocyanine Fe(II) (FePc-NHTs). Electro-spray ionization was used in conjunction with time-of-flight mass spectrometry (TOF MS). The monomer ionized as m/z where z=1 or 2, and the mass of the parent peak was 1921.25 g/mol.
- (10) FIG. 9 illustrates ¹H nuclear magnetic resonance (NMR) (600 M, 298 K, deuterated chloroform (CDCl₃)) spectrum of 2,3,9,10,16,17,23,24-octatosylaminophthalocyanine Co(II) (CoPc-NHTs).
- (11) FIG. 10 illustrates ¹³C NMR (150 M, 298 K, CDCl₃) spectrum of CoPc-NHTs.
- (12) FIG. 11 illustrates ¹H NMR (600 M, 298 K, dimethyl sulfoxide (DMSO)-d₆) spectrum of CoOAPc.
- (13) FIG. 12 illustrates high resolution mass spectroscopy analysis of 2,3,9,10,16,17,23,24-octatosylaminophthalocyanine Cu(II) (CuPc-NHTs) Electro-spray ionization was used in conjunction with TOF MS. The monomer ionized as M/1, and the mass of the parent peak was 1930.24 g/mol.
- (14) FIG. 13 illustrates powder patterns obtained for various synthetic conditions leading to FePc-Pyr-COF.

(15) FIG. 14 illustrates experimental pXRD under various conditions and simulated pXRD for CoPc-Pyr-COF. Powder patterns plotted against the computationally modeled eclipsed stacking structure.

(16) FIG. 15 illustrates powder patterns plotted against the computationally modeled eclipsed stacking structure.

(17) FIG. 16 illustrates SEM of CoPc-Pyr-COF synthesized using condition 10.

(18) FIG. 17 illustrates SEM of CuPc-Pyr-COF synthesized using condition 1.

(19) FIG. 18 illustrates scanning electron microscopy of CuPc-pyr-COF synthesized using condition 2.

(20) FIGS. 19A and 19B illustrate (FIG. 19A) calculated electronic band structure (left side) and density of state (right side) for FePc-Pyr-COF and (FIG. 19B) the corresponding first Brillouin zone and high-symmetry K-points.

(21) FIGS. 20A and 2B illustrate (FIG. 20A) calculated electronic band structure (left side) and density of state (right side) for CoPc-Pyr-COF and (FIG. 20B) the corresponding first Brillouin zone and high-symmetry K-points.

(22) FIGS. 21A and 21B illustrate (FIG. 21A) calculated electronic band structure (left side) and density of state (right side) for CuPc-Pyr-COF and (FIG. 21B) the corresponding first Brillouin zone and high-symmetry K-points.

DETAILED DESCRIPTION

(23) It is to be understood that both the foregoing general description and the following detailed description are illustrative and explanatory, and are not restrictive of the subject matter, as claimed. In this application, the use of the singular includes the plural, the word “a” or “an” means “at least one”, and the use of “or” means “and/or”, unless specifically stated otherwise. Furthermore, the use of the term “including”, as well as other forms, such as “includes” and “included”, is not limiting. Also, terms such as “element” or “component” encompass both elements or components comprising one unit and elements or components that include more than one unit unless specifically stated otherwise.

(24) The section headings used herein are for organizational purposes and are not to be construed as limiting the subject matter described. All documents, or portions of documents, cited in this application, including, but not limited to, patents, patent applications, articles, books, and treatises, are hereby expressly incorporated herein by reference in their entirety for any purpose. In the event that one or more of the incorporated literature and similar materials defines a term in a manner that contradicts the definition of that term in this application, this application controls.

(25) The development of chemically robust, porous, and electrically conductive nanomaterials drives progress in electronic devices, energy storage, catalysis, and chemical sensing. Recent advances in the synthesis of electrically conductive metal-organic frameworks (MOFs) have enabled a range of applications in electrocatalysis, energy storage, and chemical sensing that were previously believed to be inaccessible using traditionally insulating MOFs.

(26) Both through-bond and through-space charge transport mechanisms have proven to be effective in promoting conductivity in MOFs. In particular, the molecular design strategy focusing on planar two-dimensional MOFs, in which the formation of π -d conjugated sheets can promote the delocalization of charge, has yielded metallic conductivities. Despite the progress in the development of conductive MOFs, the design and synthesis of electrically conductive covalent-organic frameworks (COFs)—which arguably possess superior chemical stability—has remained a tremendous challenge. Although doping of COFs with oxidants and guest molecules has led to conductivities of $\sim 10^{-2}$ to 10^{-1} siemens per meter (S/m), the access to intrinsically conductive COFs with high conductivity remains limited.

(27) Building on the general principles of molecular engineering for other classes of conductive materials (e.g., conductive organic polymers, conductive organic molecular solids, and conductive coordination polymers), the strategy of molecular design of intrinsic conductive COFs can leverage

two established approaches: maximizing through-bond and through-space charge transport. To achieve through-bond charge transport, the covalent linkages formed during COF synthesis should be able to promote the charge delocalization efficiently.

(28) Previous theoretical and experimental studies on one-dimensional (1D) and two-dimensional (2D) conjugated polymers with single bonds in their backbones rarely showed high charge carrier mobility values, suggesting that the borate and imine linkages in typical COFs may be inefficient at facilitating through-bond charge transport. Full annulation of building blocks through aromatic linkages has been established as a promising strategy for conjugated 2D COF formation, but has not yet yielded materials with high intrinsic conductivity.

(29) To achieve through-space transport, the strategic choice of building blocks is required to maximize orbital interactions within the resulting framework structure. Although several reports have taken advantage of π -stacking for designing COF-based materials with reasonable charge carrier mobilities (8.1 cm² V⁻¹ s⁻¹), their bulk conductivities remained limited.

(30) Accordingly, a need exists for the development and use of covalent-organic frameworks with improved conductivity, sensitivity, and stability. Various embodiments of the present disclosure address the aforementioned need.

(31) In some embodiments, the present disclosure pertains to methods of sensing an analyte in a sample. In some embodiments illustrated in FIG. 1A, the methods of the present disclosure include one or more of the following steps: exposing a sample to an electrode that includes a covalent-organic framework with a plurality of metal-coordinated aromatic units linked to one another by aromatic linkers (step 10); detecting a change in a property of the electrode (step 12); and correlating the change in the property of the electrode to the presence or absence of the analyte (step 14). In some embodiments, the methods of the present disclosure can be repeated multiple times.

(32) Additional embodiments of the present disclosure pertain to covalent-organic frameworks that include various configurations. In some embodiments illustrated in FIG. 1B, the covalent-organic frameworks of the present disclosure are illustrated as covalent-organic framework 20, which includes a plurality of metal-coordinated aromatic units 22 linked to one another by aromatic linkers 24.

(33) In some embodiments, the covalent-organic frameworks of the present disclosure may be associated with electrodes. For instance, in some embodiments that are further illustrated in FIG. 1B, covalent-organic framework 20 is associated with surface 28 of electrode 26. In some embodiments, electrode 26 is a component of a device, such as a chemiresistive device.

(34) Further embodiments of the present disclosure pertain to methods of making the covalent-organic frameworks of the present disclosure. In some embodiments, such methods involve the conjugation of metal-coordinated aromatic units through the utilization of aromatic linkers.

(35) As set forth in more detail herein, the methods and covalent-organic frameworks of the present disclosure can have numerous embodiments. In particular, various covalent-organic frameworks may be utilized in various manners to sense the presence or absence of various analytes in various samples. Various methods may also be utilized to make the covalent-organic frameworks of the present disclosure.

(36) Methods of Sensing Analytes

(37) The analyte sensing methods of the present disclosure generally include one or more of the following steps: (1) exposing a sample to an electrode that includes a covalent-organic framework with a plurality of metal-coordinated aromatic units that are linked to one another by aromatic linkers; (2) detecting a change in a property of the electrode; and (3) correlating the change in the property to the presence or absence of the analyte.

(38) As set forth in more detail herein, the methods of the present disclosure can utilize various covalent-organic frameworks having various metal-coordinated aromatic units, metals, aromatic linkers, structures, and properties. Additionally, the methods of the present disclosure can be

utilized to sense various analytes from various samples.

(39) Metal-Coordinated Aromatic Units, Metals, and Aromatic Linkers

(40) The covalent-organic frameworks of the present disclosure can include various metal-coordinated aromatic units, metals, and aromatic linkers. For instance, in some embodiments, the metal-coordinated aromatic units include, without limitation, metallophthalocyanine units. In some embodiments, the metallophthalocyanine units can include, without limitation, octatosylaminophthalocyanine.

(41) The metal-coordinated aromatic units can include one or more metals. In some embodiments, the one or more metals include, without limitation, first row transition metals, divalent metals, transition metals, iron, nickel, copper, cobalt, zinc, manganese, platinum, palladium, gold, bismuth, chromium, magnesium, tin, and combinations thereof. In some embodiments, the one or more metals include, without limitation, cobalt, iron, nickel, copper, and combinations thereof.

(42) The metals of the covalent organic frameworks of the present disclosure may be in various forms. For instance, in some embodiments, the metals may be in the form of metal ions, metal clusters, metallic nodes, metal catecholates, metal salts, and combinations thereof.

(43) In some embodiments, only one type of metal may be used within the same covalent-organic frameworks. In some embodiments, more than one type of metal may be used within the same covalent-organic frameworks.

(44) The covalent-organic frameworks of the present disclosure can also include various types of aromatic linkers. For instance, in some embodiments, the aromatic linkers include, without limitation, pyrenes. In some embodiments, the pyrenes include, without limitation, pyrenetetraone.

(45) Aromatic linkers and metal-coordinated aromatic units can be linked in various manners to form covalent-organic frameworks. For instance, in some embodiments, the aromatic linkers and the metal-coordinated aromatic units are linked through pyrazine rings. In some embodiments, the pyrazine rings include, without limitation, phenazine.

(46) Analytes and Samples

(47) As set forth in more detail herein, the methods of the present disclosure can sense various analytes in various states in numerous samples in various forms. For instance, in some embodiments, the methods of the present disclosure can sense analytes in a state that include, without limitation, a gaseous state, a liquid state, and combinations thereof. In some embodiments, the analytes are in a liquid state. In some embodiments, the analytes are in a gaseous state.

(48) In some embodiments, the analytes are gaseous analytes. In some embodiments, the gaseous analytes include, without limitation, gaseous emissions, gaseous metabolites, gaseous biological signaling molecules, and combinations thereof. In some embodiments, the gaseous analytes include, without limitation, NH_3 , H_2S , NO , NO_2 , and combinations thereof.

(49) In some embodiments, the analytes are ions. In some embodiments, the ions include, without limitation, anions, cations, and combinations thereof. In some embodiments, the ions include, without limitation, Ca^{2+} , Co^{2+} , Mg^{2+} , Zn^{2+} , Pb^{2+} , Ni^{2+} , Cu^{2+} , Na^+ , K^+ , NH_4^+ , CH_3COO^- , NO_3^- , ClO_4^- , Cl^- , I^- , F^- , Br^- , SO_4^{2-} , $\text{S}_2\text{O}_8^{2-}$, $\text{Cr}_2\text{O}_7^{2-}$, and combinations thereof.

(50) In some embodiments, the analytes are present in a sample. In some embodiments, the sample can be in various forms. Such forms can include, without limitation, a gaseous form, a liquid form, a solid form, and combinations of such forms.

(51) Exposing, Detecting, and Sensing Analytes

(52) As set forth in more detail herein, the methods of the present disclosure can include various steps and means of exposing an analyte in a sample to an electrode that includes the covalent-organic frameworks of the present disclosure. For instance, in some embodiments, the exposing results in the reversible association of any analyte in the sample with the covalent-organic framework. In some embodiments, the exposing results in the selective association of a specific

analyte in the sample with the covalent-organic framework.

(53) Various methods may also be utilized to detect a change in a property of an electrode that includes the covalent-organic frameworks of the present disclosure. For instance, in some embodiments, the detected change in the property of the electrode is a change in normalized conductance. In some embodiments, the change in normalized conductance is characterized by the following formula: $-\Delta G/G_{sub.0} = -(I_{sub.0} - I)/I_{sub.0} - 100\%$, where $-\Delta G/G_{sub.0}$ represents normalized conductance, $I_{sub.0}$ represents initial current, and I represents current at various points during measurement.

(54) In some embodiments, the detected change in the property of the electrode is a change in the potential of the electrode. In some embodiments, the change in the potential of the electrode is detected by sensing a change in voltage of the electrode over time.

(55) Various methods may also be utilized to correlate a change in a property of an electrode to the presence or absence of an analyte in a sample. For instance, in some embodiments, the change in a property of the electrode is correlated to the presence or absence of an analyte by comparing the change in the property to the change in property of the electrode in response to known analytes. In some embodiments, the change in a potential of the electrode is correlated to the presence or absence of an analyte by comparing the change in the potential to potentiometric slopes of known analytes.

(56) In some embodiments, the sensing of analytes only includes detecting the presence or absence of the analyte in a sample. In additional embodiments, the sensing also includes detecting the concentration of the analyte.

(57) In some embodiments, the methods of the present disclosure can sense analytes in a sample at a low concentration. For instance, in some embodiments, the methods of the present disclosure can sense analytes at concentrations of less than 500 parts-per-billion (ppb). In some embodiments, the methods of the present disclosure can sense analytes at concentrations of less than 250 ppb. In some embodiments, the methods of the present disclosure can sense analytes at concentrations of less than 100 ppb. In some embodiments, the methods of the present disclosure can sense analytes at concentrations of less than 75 ppb. In some embodiments, the methods of the present disclosure can sense analytes at concentrations of less than 50 ppb. In some embodiments, the methods of the present disclosure can sense analytes at concentrations of less than 25 ppb. In some embodiments, the methods of the present disclosure can sense analytes at concentrations of less than 10 ppb. In some embodiments, the methods of the present disclosure can sense analytes at concentrations of less than 5 ppb.

(58) Covalent-Organic Framework Structures

(59) The covalent-organic frameworks of the present disclosure can have numerous structures. For instance, in some embodiments, the covalent-organic frameworks of the present disclosure are in the form of a fully aromatic and conjugated structure. In some embodiments, the covalent-organic frameworks are in the form of a two-dimensional (2D) conductive network. In some embodiments, the covalent-organic frameworks are in a crystalline form. In some embodiments, the covalent-organic frameworks are in the form of a layered structure.

(60) In some embodiments, the covalent organic frameworks of the present disclosure exhibit a nanosheet morphology with regular and/or parallel lines. In some embodiments, the covalent organic frameworks exhibit nanoribbon structures with grids and/or parallel lines.

(61) In some embodiments, the covalent-organic frameworks of the present disclosure are in the form of planar sheets. In some embodiments, the covalent-organic frameworks are in the form of rods (e.g., rods with lengths of several micrometers and widths of hundreds of nanometers). In some embodiments, the covalent-organic frameworks have a square-like cage structure. In some embodiments, the covalent-organic frameworks have lattices with Lieb topology.

(62) In some embodiments, the covalent-organic frameworks of the present disclosure are porous structures. In some embodiments, the covalent-organic frameworks of the present disclosure

include a plurality of micropores. In some embodiments, pores of the porous structures include lengths ranging from about 1.0 nm to about 5.0 nm. In some embodiments, the pores have lengths ranging from about 2.0 nm to about 3.0 nm. In some embodiments, the pores have lengths ranging from about 1.0 nm to about 2.5 nm. In some embodiments, the pores have lengths ranging from about 2.0 nm to about 2.5 nm. In some embodiments, the pores have lengths of about 2.17 nm. (63) The pores of the covalent organic frameworks of the present disclosure can have various shapes. For instance, in some embodiments, the pores of the covalent-organic frameworks are in the form of square apertures.

(64) Covalent-Organic Framework Properties

(65) The covalent-organic frameworks of the present disclosure can have various advantageous properties. For instance, in some embodiments, the covalent-organic frameworks of the present disclosure exhibit high conductivities. In some embodiments, the covalent-organic frameworks of the present disclosure have bulk conductivities higher than 1×10^{-2} S/m. In some embodiments, the covalent-organic frameworks of the present disclosure have bulk conductivities higher than 1×10^{-3} S/m. In some embodiments, the covalent-organic frameworks of the present disclosure have bulk conductivities higher than 2×10^{-3} S/m. In some embodiments, the covalent-organic frameworks of the present disclosure have bulk conductivities of at least 2.5×10^{-3} S/m.

(66) Electronic Devices

(67) The covalent-organic works of the present disclosure may be utilized as various components of various electronic devices. For instance, in some embodiments, the covalent-organic frameworks can be utilized as components of an electrode. In some embodiments, the electrode includes, without limitation, carbon electrodes, glassy carbon electrodes, gold electrodes, solid contact electrodes, and combinations thereof.

(68) In some embodiments, the covalent-organic frameworks of the present disclosure are positioned on an electrode surface. In some embodiments, the covalent-organic frameworks serve as the electrode surface.

(69) In some embodiments, the electrodes that contain the covalent-organic frameworks of the present disclosure also include additional components. Such additional components can include, without limitation, wirings, a potentiostat, an output display, and combinations of such components. In some embodiments, the wiring electrically connects the electrode surface to the potentiostat. In some embodiments, the output display is electrically connected to the potentiostat.

(70) Electrodes that contain the covalent-organic frameworks of the present disclosure can be associated with various electronic devices. For instance, in some embodiments, the electrodes are associated with chemiresistive devices.

(71) Methods of Making Covalent-Organic Frameworks

(72) Additional embodiments of the present disclosure pertain to methods of making the covalent-organic frameworks of the present disclosure. In some embodiments, such methods involve the conjugation of metal-coordinated aromatic units through the utilization of aromatic linkers. In some embodiments, the conjugation occurs through aromatic annulation. In some embodiments, the conjugations results in the formation of pyrazine rings.

(73) In some embodiments, the conjugation occurs by mixing metal-coordinated aromatic units and aromatic linkers in an acidic solution. In some embodiments, the acidic solution contains sulfuric acid.

(74) In some embodiments, the mixing of metal-coordinated aromatic units and aromatic linkers also involves heating. For instance, in some embodiments, the mixing occurs at temperatures above 100° C. In some embodiments, the mixing occurs at temperatures above 150° C. In some embodiments, the mixing occurs at temperatures above 200° C.

(75) In some embodiments, the methods of the present disclosure also include a step of associating the formed covalent-organic frameworks of the present disclosure with a component of an

electronic device. For instance, in some embodiments, the formed covalent-organic frameworks may be associated with a surface of an electrode.

(76) Various methods may be utilized to associate covalent-organic frameworks with an electronic device component. For instance, in some embodiments, the covalent-organic frameworks of the present disclosure are associated with an electronic device component by methods that include, without limitation, drop casting, sputtering, spraying, and combinations thereof.

(77) In some embodiments, the covalent-organic frameworks of the present disclosure can be formed at the same time as their association with an electronic device component. For instance, in some embodiments, the conjugation of metal-coordinated aromatic units through the utilization of aromatic linkers can occur on a surface of an electrode material in order to form chemiresistive devices.

(78) Applications and Advantages

(79) In some embodiments, the present disclosure describes the development of compositionally novel and intrinsically conductive covalent organic frameworks through the full aromatic annulation of metal-coordinated aromatic units (e.g., octaamino-nickelphthalocyanine) and aromatic linkers (e.g., pyrenetetraone). In some embodiments, the porous and conductive attributes of the covalent organic frameworks of the present disclosure exhibit optimal chemiresistive responses to gases, including NH_3 , H_2S , NO , and NO_2 .

(80) The present disclosure provides at least three important advancements in the development of conductive and two-dimensional porous materials. First, the covalent organic frameworks of the present disclosure provide a compositionally novel material which links metal-coordinated aromatic units (e.g., metallophthalocyanine units) within planar two-dimensional sheets using extremely robust and chemically stable aromatic linkers (e.g., conjugated linkers, such as pyrene). In some embodiments, this unique modular design yields a bulk conductivity of $2.51 \times 10^{-3} \text{ S/m}$, which constitutes the highest value obtained for an intrinsically conductive covalent organic framework.

(81) Second, Applicants demonstrate the first use of covalent organic framework-based chemiresistive electronic devices for detection of various reducing and oxidizing gases. For instance, as described in more detail in the Examples herein, Applicants have demonstrated that the chemiresistive devices of the present disclosure can show promising responses for gaseous analyte sensing, including NH_3 and NO .

(82) Third, Applicants envision that the covalent organic frameworks of the present disclosure will offer access to multi-functional properties (e.g., porosity, conductivity, magnetism, and catalytic ability) within an extremely stable chemical scaffold. For instance, Applicants envision that embedded metal-coordinated aromatic units (e.g., metallophthalocyanine units) with their modular metal center can provide the possibility to develop a versatile platform to tune the surface chemistry, conductivity and magnetic properties of the resulting materials. Moreover, Applicants envision that the chemical stability of the covalent organic frameworks of the present disclosure will provide significant advantages in material stability under harsh chemical conditions.

(83) Additional Embodiments

(84) Reference will now be made to more specific embodiments of the present disclosure and experimental results that provide support for such embodiments. However, Applicants note that the disclosure below is for illustrative purposes only and is not intended to limit the scope of the claimed subject matter in any way.

EXAMPLE 1

Two-Dimensional Chemiresistive Covalent Organic Framework with High Intrinsic Conductivity

(85) This Example describes the synthesis of a novel intrinsically conductive two-dimensional (2D) covalent organic framework (COF) through the aromatic annulation of 2,3,9,10,16,17,23,24-octaaminophthalocyanine nickel(II) and pyrene-4,5,9,10-tetraone. The intrinsic bulk conductivity of the COF material (termed COF-DC-8) reached $2.51 \times 10^{-3} \text{ S/m}$, and increased by three orders of

magnitude with I.sub.2 doping. Electronic calculations revealed an anisotropic band structure, with the possibility for significant contribution from out-of-plane charge-transport to the intrinsic bulk conductivity. Upon integration into chemiresistive devices, this conductive COF showed excellent responses to various reducing and oxidizing gases, including NH.sub.3, H.sub.2S, NO, and NO.sub.2, with parts-per-billion (ppb) level of limits of detection (LOD for NH.sub.3=70 ppb, for H.sub.2S=204 ppb, for NO=5 ppb, and for NO.sub.2=16 ppb based on 1.5 min exposure). Electron paramagnetic resonance spectroscopy and X-ray photoelectron spectroscopy studies suggested that the chemiresistive response of the COF-DC-8 involves charge transfer interactions between the analyte and nickelphthalocyanine component of the framework.

EXAMPLE 1.1

Introduction

(86) The development of chemically robust, porous, and electrically conductive nanomaterials drives progress in electronic devices, energy storage, catalysis, and chemical sensing. Recent advances in the synthesis of electrically conductive metal-organic frameworks (MOFs) have enabled a range of applications in electrocatalysis, energy storage, and chemical sensing that were previously inaccessible using traditionally insulating MOFs. Both through-bond and through-space charge transport mechanisms have proven to be effective in promoting conductivity in MOFs. In particular, the molecular design strategy focusing on planar two-dimensional MOFs, in which the formation of π -d conjugated sheets can promote the delocalization of charge, has yielded metallic conductivities. Despite progress in the development of conductive MOFs, the design and synthesis of electrically conductive π -conjugated covalent organic frameworks (COFs) connected by chemically robust bonds—which arguably possess superior chemical stability to frameworks derived from reversible coordination chemistry and borate or Schiff-base chemistry—has remained a tremendous challenge. Although doping of COFs with oxidants and guest molecules has led to conductivities of $\sim 10^{-2}$ to 10^{-1} S/m, access to intrinsically conductive COFs with high bulk conductivity remains limited.

(87) Capitalizing on the general principles of molecular engineering for other classes of conductive materials (e.g., conductive organic polymers, conductive organic molecular solids, and conductive coordination polymers), the strategy for maximizing the intrinsic bulk conductivity of COFs can leverage through-bond and through-space charge transport characteristics. To achieve through-bond charge transport, the covalent linkages formed during COF synthesis must promote efficient charge delocalization. Previous studies on one-dimensional (1D) and two-dimensional (2D) conjugated polymers with single bonds in their backbones rarely showed high charge carrier mobility values, suggesting that borate and imine linkages in typical COFs may be inefficient at facilitating through-bond charge transport.

(88) While full annulation of building blocks through aromatic linkages has been established as a promising strategy for conjugated 2D COF formation, it has not yet yielded materials with high intrinsic conductivities. Interestingly, recent studies highlighted the unique role of topology in enhancing through-bond charge delocalization in conjugated 2D COFs, with Lieb lattice being more favorable over Kagome lattice. A complementary strategy for achieving through-space transport requires maximizing orbital interactions within the resulting layered framework structure through strategic choice of building blocks.

(89) Although several reports have taken advantage of π -stacking for designing COF-based materials with reasonable charge carrier mobilities ($8.1 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$), their bulk conductivities remained limited. Applicants reasoned that by simultaneously capitalizing on both molecular design strategies of through-bond and through-space transport in a highly planar, two-dimensional, fully conjugated material based on a Lieb lattice may lead to the realization of a high intrinsic bulk electrical conductivity within a COF-based material.

(90) This Example describes the development of a novel intrinsically conductive COF (COF-DC-8) through the condensation reaction between the highly planar conjugated building blocks:

octaamino-derived nickelphthalocyanine (NiOAPc) and pyrenetetraone (TOPyr). The annulation of tetraketone and octaamine precursors to form pyrazine rings generates a fully aromatic conjugated framework structure with square apertures (FIG. 2). The utilization of the nickelphthalocyanine core was inspired by its application in the construction of conductive framework materials demonstrated by Applicants' group and others, as well as its affinity to small molecules. The resulting fully conjugated 2D lattice with Lieb topology can maximize through-bond charge delocalization, while the stacking of the embedded metallophthalocyanine units confined within the rigid framework material can efficiently promote out-of-plane charge transfer. The bulk conductivity—which characterizes the weighted average of contributions from through-bond and through-space charge delocalization in polycrystalline COF-DC-8—reached 2.51×10^{-3} S/m, representing the highest bulk conductivity achieved within an intrinsically conductive COF.

(91) Chemiresistive devices made from this conductive COF showed excellent responses and ultra-low limits of detection for gaseous analytes (1.5 min exposure-based LODs: 70 ppb for NH₃, 204 ppb for H₂S, 5 ppb for NO, and 16 ppb for NO₂). Increases in resistance towards reducing gases and decreases in resistance towards oxidizing gases were consistent with p-type semiconductive character of the COF.

(92) Spectroscopic characterization using electron paramagnetic resonance (EPR) and X-ray photoelectron spectroscopy (XPS) suggested that the chemiresistive response of COF-DC-8 originates from the combination of binding and charge transfer interactions between the analyte and the Ni-containing phthalocyanine component of the framework. The promising chemiresistive performance highlights the potential application of this modular class of materials in the fabrication of electronic devices and chemical sensors.

EXAMPLE 1.2

Synthesis and Characterization

(93) As illustrated in FIG. 2, the condensation reaction performed in a mixed solvent of dimethylacetamide (DMAC) and o-dichlorobenzene (DCB) in the presence of sulfuric acid for 10 days gave the desired COF-DC-8 as a dark green powder. Microwave heating significantly reduced the reaction time to 10 hours, albeit at the expense of slightly diminished crystallinity. Efforts to optimize reaction conditions yielded several alternative approaches for accessing COF-DC-8, including the use of the aqueous solution of acetic acid in a mixed solvent system of DMAC and DCB, and the use of NMP as the solvent in the presence of sulfuric acid or aqueous solution of acetic acid.

(94) Fourier-transform infrared spectroscopy of COF-DC-8 showed the appearance of characteristic absorption bands of the phenazine system at 1518, 1431, and 1351 cm⁻¹, while absorption bands of C=O and —NH₂ groups from NiOAPc and TOPyr were absent, indicating the formation of phenazine linkages. XPS showed characteristic bands for the K-edge of carbon (285.9 eV) and nitrogen (398.6 eV) in C=N bond, indicating the presence of sp²-hybridized nitrogen atoms. Elemental analysis showed that C:N:Ni ratios were consistent with theoretical values, although the absolute percentages of C, N, and Ni were slightly lower than theoretical values. The discrepancy between the theoretical and observed values could be from the existence of unreacted C=O and NH₂ groups at sheet edges and the entrapment of small-molecule volatiles (acetone, H₂O, or DMAC) included in the pores, as suggested by XPS and TGA analysis.

(95) Powder X-ray diffractometry (PXRD) further supported the formation of the desired framework formed under optimized reaction conditions (FIGS. 3A-3B). Prominent peaks were observed at $2\theta = 4.06^\circ$, 8.19° , 11.48° , 12.26° , and 26.65° , that were assigned to the [100], [200], [220], [300], and [001] facets, respectively (FIG. 3B). Although PXRD displayed peak broadening which was possibly due to small crystallite size, the peak intensity and position were sufficient for confirming the framework structure and key cell parameters of DC-COF-8. The strong diffractions from the [100] and [200] facets indicated long-range order within the ab plane of the structure. The

Ni-to-Ni distance on the side of the square, based on those two diffractions, was calculated to be 2.17 nm. The presence of the [001] facet at 26.65° suggested the structural ordering with a 3.34 Å separation of layers along the c axis perpendicular to the 2D sheets. These results were consistent with the structure optimized by DFT calculations with generalized gradient approximation functional. The experimental PXRD was in good agreement with the simulated fully eclipsed AA-stacking model with a space group of P4/MMM characterized by cofacial stacking of metallophthalocyanine units in adjacent layers; in contrast, the staggered model with an offset by distances of $a/2$ and $b/2$ did not match the experimental PXRD pattern (FIG. 4B).

(96) Scanning electron microscopy (SEM) revealed the presence of rod-shaped crystallites with lengths ranging from hundreds of nanometers to several micrometers and widths of hundreds of nanometers (FIG. 3C). Transmission electron microscopy (HR-TEM) provided visualization of a layered morphology (FIG. 3D). The presence of regular lines with a spacing of ~ 2.2 nm in TEM (FIG. 3D) was consistent with the interatomic distances of the [100] plane inferred from PXRD and computational models. COF-DC-8 exhibited reversible nitrogen sorption isotherm curves with a Brunauer-Emmett-Teller surface area of $360 \text{ m}^2 \text{ g}^{-1}$. The pore-size distribution profile gave an accessible pore size of 1.7 nm. Thermogravimetric analysis showed excellent thermal stability of COF-DC-8 with only a total 10% of mass loss upon heating to 510°C . COF-DC-8 maintained its crystallinity when treated with 12 M HCl and 14 M KOH for 24 h, confirming its chemically robust skeleton.

EXAMPLE 1.3

Electronic Properties

(97) The electrical conductivity of COF-DC-8 was measured using a four-point probe method on a pressed pellet with a thickness of 1.0 mm. The bulk conductivity of COF-DC-8 was determined to be $2.51 \times 10^{-3} \text{ S/m}$ at 298 K, which is several orders of magnitude higher than small-molecule nickel phthalocyanine and a large number of other organic solids, and is also significantly higher than many undoped conjugated polymers (10^{-4} – 10^{-8} S/m) and other pristine semi-conductive COFs. The Arrhenius fit to variable-temperature current measurements revealed an activation energy of 324 meV for the transport of charge carriers in COF-DC-8. Nearly a 10³-fold improvement in conductivity was achieved by doping the COF with a gaseous stream of 12 ($\sim 400 \text{ ppm}$ in N_2) within 5 minutes of exposure at room temperature.

(98) DFT calculations were performed to gain insight into the electronic structure of the COF. The Dirac bands crossed the Fermi level in both A-M, Γ -Z, and R-X directions and exhibited wide band dispersions of more than 0.5 eV, suggesting the intrinsically conductive nature of COF-DC-8. The high symmetry points in the first Brillouin zone of COF-DC-8 showed that the bands crossed the Fermi level through the out-of-plane directions (FIG. 4A). In contrast, the in-plane direction, including Γ -M, Z-R, Γ -M and Z-A of the material, exhibited moderate band gaps ranging from 0.5–1.0 eV. These results suggested that the charge transport in COF-DC-8 may be anisotropic and that a significant mechanism contributing to the bulk conductivity may be through pathways along the c-axis. The computed through space charge-transport property is consistent with those found in other phthalocyanine- and porphyrin-based COFs, that exhibited high carrier mobility along the direction of the stacking due to the formation of periodic π -columns. Applicants hypothesize that the eclipsed stacking mode of COF-DC-8 has the potential to amplify the alignment of the π -conjugated units and enhance the through-space metallophthalocyanine-to-metallophthalocyanine charge transport along the 2D stacks. Since the experimental validation of these calculations would require access to highly oriented nanosheets and nanowires of the COF, the polycrystalline material—which was the focus of this Example—was not suitable for confirming these calculations experimentally.

EXAMPLE 1.4

Chemiresistivity of COF-DC-8

(99) 2D COFs are highly appealing for the fabrication of electronic devices by virtue of their well-

defined lattice with π functionality, large surface-to-volume ratio, and chemically robust skeleton. However, the poor to modest conductivity of existing 2D COFs has limited their use in electronic devices and other applications. Given the good electrical conductivity and the ability of the built-in nickelphthalocyanine units to bind a wide range of analytes through coordination with the metal center, Applicants tested the chemiresistive response of COF-DC-8 towards a series of gases with different reducing and oxidizing abilities, including NH_3 , H_2S , NO , and NO_2 . These gases constitute well-known toxic pollutants and are also biological signaling molecules. To make chemiresistive devices, a COF suspension was drop casted onto interdigitated electrodes, which gave devices with resistances in the range of $\text{k}\Omega$. The devices were tested under a constant applied voltage of 1.0 V, and the current was monitored using a potentiostat.

(100) Upon exposure to 40 ppm of NH_3 (balance gas N_2) for 30 min, COF-DC-8 showed a normalized response ($-\Delta G/G_0$) of $39 \pm 1\%$ (FIG. 5A). This positive change in resistance of COF-DC-8 in response to NH_3 , which typically serves as an electron donor, suggested that COF-DC-8 behaved as a p-type semiconductor. A response of $62 \pm 1\%$ was observed to 40 ppm of H_2S (FIG. 5A). The higher response to H_2S , compared to NH_3 , may be due to the stronger interaction between the material and H_2S . This response value was comparable with chemiresistive devices employing hexahydroxytriphenylene-based MOF and metallophthalocyanine-based bimetallic MOF. In contrast to the positive responses to NH_3 and H_2S , COF-DC-8 exhibited negative response to both NO and NO_2 (FIG. 5B), which further reinforced the p-type character of COF-DC-8. The exposure of the COF to 40 ppm NO generated a high response of $3939 \pm 317\%$ (FIG. 5C). This remarkable response to NO was more than 10 times higher than that produced by triphenylene-based conductive MOFs. An even higher response of $-6338 \pm 479\%$ was generated after the 30 min exposure of 40 ppm of NO_2 (FIG. 5D). This modulation of electrical conductivity of this COF by chemical doping with low-concentrations of small reactive gases has promising implications for developing advanced electronics where rectification of conductance is required.

EXAMPLE 1.5

Determination of LODs

(101) Concentration-dependent studies (FIGS. 5A-5D) enabled quantitative correlations between changes in response as a function of analyte concentration for COF-DC-8. Although the response values were distinguishable at $[\text{NH}_3] = 2\text{-}80$ ppm, only a relatively narrow linear range of 2-10 ppm could be identified (FIG. 5E). The LOD value for NH_3 determined from the response-concentration relationship in the linear range was 70 ppb based on 1.5 min exposure. Compared with NH_3 , COF-DC-8 showed a wider linear range of 5-80 ppm for H_2S that persisted across different exposure times ranging from 1.5 min to 30 min (FIG. 5F). The LOD derived for H_2S detection was 204 ppb based on response values after 1.5 min exposure to the analyte. The calculated LOD values for both NH_3 and H_2S showed minimal dependence on exposure time with only slight changes observed for exposure time differences in the range of 1.5 min to 30 min (70-57 ppb for NH_3 , 204-121 ppb for H_2S).

(102) Excellent linear response-concentration relationships ($R^2 = 0.98\text{-}0.99$) were observed for NO and NO_2 at relatively wide concentration ranges of 0.02-40 ppm for NO and 2-40 ppm for NO_2 (FIGS. 5G, 5H). The striking sensitivity of COF-DC-8 led to ultra-low LODs of 5 and 16 ppb for NO and NO_2 , respectively, based on responses after 1.5 min exposure. Compared with those found for NH_3 and H_2S , significant exposure-time dependences were found for the LOD values of NO and NO_2 . Longer exposure times led to sub-ppb level of calculated LODs.

EXAMPLE 1.6

Recovery of Device Performance

(103) Sensors characterized by high reversibility in response under repeated exposures to analyte are advantageous for their reusability. While dosimetric devices are highly irreversible when cycled through analyte exposures, they are advantageous to allow a historical record of total exposure

events. The response of COF-DC-8 to NH₃ was partially reversible in the concentration range from 5 to 80 ppm (FIG. 5A). Full reversibility was observed when the concentration was below 2 ppm. Across the entire tested concentration range (2-80 ppm) for H₂S, only slight reversibility (5%-15%) was observed (FIG. 5B). The response to both NO and NO₂ showed partial reversibility (7%-33% for NO and 25%-45% for NO₂, FIG. 5C and FIG. 5D). These observations demonstrated the dosimetric or quasi-dosimetric characteristics of COF-DC-8 to H₂S, NO, and NO₂, in which the response was dependent on the dosing history.

(104) In order to investigate the reusability of the devices, the analyte-exposed COF devices were reactivated by both thermal recovery and solvent recovery. After solvent recovery by immersing in deionized water for 1 hour, the responses of devices were largely restored (108% for H₂S, 71% for NO, and 100% for NO₂). However, the devices were only partially restored (60% for H₂S, 14% for NO, and 17% for NO₂) their performance after thermal recovery by heating at 60° C. for 18 h under ~20 mTorr. These results were consistent with the previous report showing that, in metal-organic frameworks, washing was more efficient than heating for recovery of device performance. The recovery tests suggested that analyte molecules bound by the material may be released from the surface under aqueous conditions, thus restoring the sensing ability of the COF-DC-8 device.

EXAMPLE 1.7

Analysis of Sensing Kinetics

(105) The rate and magnitude of the chemiresistive response is usually governed by several factors, including the characteristics of the sensing apparatus, the morphology and preferential orientation of the material, and the electronic nature of the analyte. Analyzing the rate of response at the initial stage upon analyte exposure is a convenient technique that can allow rapid concentration-dependent measurements and convenient assessment of sensing kinetics when factors of the sensing apparatus and material characteristics are kept constant.

(106) Through the analysis of the initial rate of response, Applicants found that COF-DC-8 could differentiate certain analytes and their concentrations quantitatively within only 1 minute of initial analyte exposure (FIGS. 6A and 6B). Plotting the slope of initial response over the first 1 min of exposure versus the concentration of the four gases provided a linear relationship, which, in most cases, extended into much wider concentration ranges compared with those found in direct response-concentration analysis. This method provided a simple and fast analysis of the concentration of the four analytes.

(107) Each of the four gases showed distinct dynamics of response at wide concentration range. Under the concentration of 40 ppm, the rate of the initial response for NH₃ was 48±5% min.⁻¹, while the initial response for H₂S showed a slower rate of 18±3% min.⁻¹. Compared with reducing gases NH₃ and H₂S, COF-DC-8 showed much quicker response for NO and NO₂ at the initial state with rates at 513±53% min.⁻¹ and 171±8% min.⁻¹, respectively. This observation indicated that the initial rate of response was strongly dependent on type of the analyte-material interaction.

EXAMPLE 1.8

Sensing Mechanism

(108) To investigate the sensing mechanism, spectroscopic analysis using EPR and XPS was performed. EPR analysis was carried out in the X-band at 77 K under N₂ atmosphere on COF samples before and after they were exposed to analytes (1% NH₃, H₂S, NO, and NO₂) for 1 hour. XPS analysis was carried out on analyte-exposed samples prepared by a similar procedure with those for EPR analysis, which were subsequently mounted on copper tape and analyzed under reduced pressure of ~10⁻⁷ Torr.

(109) The EPR spectrum of pristine COF-DC-8 showed a ligand centered radical at g=1.99, which was likely due to the chemisorption of O₂ on the Ni center that induces unpaired radicals because of the charge transfer from the phthalocyanine ligand to O₂ molecule. After NH₃

exposure, only a 15% intensity increase in EPR signal was observed (FIG. 7A). No obvious change was detected on both the XPS spectra of C 1s, N 1s, and Ni 2p. These results suggested that the interaction between NH₃ and COF-DC-8 may be dominated by reversible and weak charge transfer interactions under tested conditions.

(110) After dosing COF-DC-8 with H₂S, Applicants observed a significant decrease in peak intensity of 70% in the EPR signal (FIG. 7A). This observation may be consistent with the replacement of surface-bound O₂ by H₂S at the Ni centers. By XPS, clear emission lines at binding energies of 163.8 eV and 165.1 eV (FIG. 7C) were detected, that were similar to S 2p_{1/2} and S 2p_{3/2} peaks found in metal sulfides, which indicated the formation of Ni-S bond. Moreover, a prominent peak at a higher binding energy of 167.5 eV, consistent with the formation of sulphite species, was also observed. This observation suggested that the Ni-bound H₂S may be further transformed into species with higher oxidation states, likely with the participation of surface adsorbed O₂ through a catalytic oxidation reaction. This type of irreversible transformation is consistent with the negligible recovery in response after H₂S exposure shown in FIG. 5B.

(111) After the exposure of the COF to oxidizing gases, prominent increases of 3.2- and 4.2-fold in the EPR signals were found after NO and NO₂ exposure, respectively (FIG. 7B). No significant changes were observed for the XPS spectra of Ni 2p, indicating that the oxidation state of the Ni centers was not altered. After NO dosing, a new component corresponding to N is appeared at the binding energy of 402.0 eV, which was absent in the pristine material. Based on the previous studies of NO adsorption on metal surfaces, this component was assigned as a Ni-bound NO species. After NO₂ dosing, a weak, yet clear, peak was observed at the binding energy of 406.1 eV (FIG. 7D). Through the comparison with spectra of NO₂ adsorbed on metallophthalocyanines and metal oxides, this peak suggested the existence of NO₂ molecules which were likely coordinated to the Ni center. Taken together, the EPR and XPS results suggest that surface adsorption and charge transfer interactions between the analytes and nickelphthalocyanine component of the COF were related to the observed chemiresistive properties of the COF.

EXAMPLE 1.9

Conclusion

(112) In conclusion, Applicants have designed and synthesized a novel intrinsically conductive COF-DC-8 through the aromatic annulation between of 2,3,9,10,16,17,23,24-octa-aminophthalocyanine nickel(II) and pyrene-4,5,9,10-tetraone. The reaction between the tetraketone and octaamine allows the formation of pyrazine rings to generate a fully aromatic conjugated structure, and renders the construction of square apertures with a side length of 2.2 nm and excellent chemical and thermal stability. The bulk room temperature conductivity of COF-DC-8 reached 2.51×10^{-3} S/m, which is the highest bulk conductivity achieved within an intrinsically conductive COF. Doping with I₂ can further increase the bulk conductivity by a factor of 10^3 .

(113) In the first implementation of a conductive COF as active material for chemiresistive devices, COF-DC-8 showed excellent responses toward NH₃, H₂S, NO, and NO₂ with ultra-low limits of detection at ppb level, that is 70 ppb for NH₃, 204 ppb for H₂S, 5 ppb for NO, and 16 pb for NO₂ after only 1.5 min exposure to the analytes. These values are comparable to or superior to the most sensitive sensors based on conductive MOFs. The sensing performance also significantly surpasses that of small-molecule metallophthalocyanines based chemiresistors, considering the high sensitivity and low driving voltage demand of COF-DC-8.

EXAMPLE 2

The Synthesis of MPc-Containing Covalent-Organic Frameworks and Their Applications in Chemical Sensing

(114) This Example illustrates the synthesis of additional metallophthalocyanine (MPc)-containing

covalent-organic frameworks and their applications in chemical sensing.

(115) Chemicals were purchased from Sigma Aldrich or TCI and used as received. High-Resolution MS was performed at the University of Illinois Urbana-Champaign. NMR spectra were recorded on Bruker 500 and 600 MHz NMR spectrometer. The chemical shifts (δ) were expressed in ppm with internal standard tetramethylsilane (TMS) and solvent signals as internal references, and J values were given in Hz. Standard abbreviations indicating multiplicity were used as follows: s (singlet), br (broad), d (doublet), t (triplet), q (quartet), m (multiplet).

(116) Scanning Electron Microscopy (SEM) and Energy Dispersive X-ray Spectroscopy (EDX) were performed using a Hitachi TM3000 SEM (Tokyo, Japan) equipped for X-ray microanalysis with a Bruker Edax light element Si(Li) detector (Billerica, MA). Transmission electron microscopy was carried out at a Tecnai F20ST FEG TEM instrument. Powder X-ray diffraction (PXRD) measurements were performed with a Rigaku sixth generation MiniFlex X-ray diffractometer with a 600 W (40 kV, 15 mA) CuK α (λ =1.54 Å) radiation source.

EXAMPLE 2.1. General Synthesis of MPc-Pyr-COF (M=Fe, Co, Cu)

(117) Synthesis of the MOAPc ligand (M=Fe, Co, Cu). The synthetic scheme toward FeOAPc ligand is shown below in Scheme 1.

(118) ##STR00001##

(119) Synthesis of 2,3,9,10,16,17,23,24-octatosylaminophthalocyanine Fe(II) (FePc-NHTs). The synthesis of the 2,3,9,10,16,17,23,24-octatosylaminophthalocyanine Ni(II) was adapted from the literature. To a suspension of 1.00 g (2.1 mmol) of N,N'-(4,5-dicyano-1,2-phenylene)bis(4-methylbenzenesulfonamide) and 1.12 g of anhydrous FeCl₃·2H₂O (2.1 mmol, 1.0 eq) in n-hexanol (3 mL) in a glass pressure vessel, 2 mL of DBU (1,8-Diazabicyclo[5.4.0]undec-7-ene) was added. The system was purged with N₂ for 10 minutes and then sealed. The reaction vessel was then heated at 170° C. for 36 hours. The reaction was then allowed to cool to room temperature. Once at room temperature, the reaction mixture was dissolved in 150 mL of a mixed solvent system of acetic acid and CH₂Cl₂ (v/v=1:5). The organic extract was washed with water (200 mL×3) and then evaporated to dryness to provide an emerald-colored oil. The residual hexanol remaining in the oil was removed by repeatedly dispersing in 40 mL of a mixture CH₂Cl₂/petroleum ethers (v/v=1:3) and decanting the brownish solvent. The emerald-colored crude product was dissolved in CH₂Cl₂ and pyridine and filtered through a fine fitted funnel. The organic mother liquor was evaporated to dryness and re-dissolved in 20 mL MeOH. The pyridine was quenched with conc. HCl (3 mL). Upon addition of HCl a solid was observed to crash out of solution. Filtration produced a fine black powder and a brown-green mother liquor. The solids were further purified by column chromatography (silica gel, eluent: CH₂Cl₂/MeOH/pyridine=50/2.5/0.2) to give product FePc-NHTs as a green solid (250 mg, yield 67%). ¹H-NMR (500 MHz, dimethyl sulfoxide (DMSO)-d₆) no proton peaks were observed in low-field NMR experiments. HRMS (ES⁺): calculated for C₈₈H₇₂FeN₁₆O₁₆S₈ 1921.97 [M⁺], 960.98 [M²⁺]; found 1921.26 M/z, 961.63 m/z.

(120) FIG. 8 illustrates high resolution mass spectroscopy analysis of FePc-NHTs. Electro-spray ionization was used in conjunction with TOF MS. The monomer ionized as m/z where z=1 or 2, and the mass of the parent peak was 1921.25 g/mol.

(121) Synthesis of FeOAPc. To a 50 mL round bottom flask charged with 47.4 mg of 2,3,9,10,16,17,23,24-octatosylaminophthalocyanine Cu(II), 142.2 μ L of deionized water and 1.42 mL of concentrated H₂SO₄ were successively added. The mixture was then heated to 110° C. for 50 minutes under N₂. The reaction was then cooled to room temperature, and the reaction mixture was poured into ice-water (25 mL). The dark green precipitate was collected by centrifugation and the solid was washed thoroughly with deionized water (20 mL×3), 10% KOH (20 mL×3), deionized water (20 mL×3), and acetone (20 mL×3). The product was obtained as a dark purple to black powder (yield: 4.4 mg, 25.7%). ¹H-NMR (500 MHz, DMSO-d₆) no

proton peaks were observed in low-field NMR experiments.

(122) Synthesis of CoOAPc. The synthetic scheme toward CoOAPc ligand is shown below in Scheme 2.

(123) ##STR00002##

(124) Synthesis of 2,3,9,10,16,17,23,24-octatosylaminophthalocyanine Co(II) (CoPc-NHTs). To a suspension of 4.67 g (10 mmol) of N,N'-(4,5-dicyano-1,2-phenylene)bis(4-methylbenzenesulfonamide) and 1.29 g of anhydrous CoCl₂·2H₂O (10 mmol, 1.0 eq) in n-hexanol (15 mL) in a glass pressure vessel, 5 mL of DBU (1,8-Diazabicyclo[5.4.0]undec-7-ene) was added. The system was purged with N₂ for 10 minutes and then sealed. The reaction vessel was then heated at 175° C. under N₂ for 60 hours. Upon cooling to room temperature, the reaction mixture was dissolved in 500 mL mixed solvent of acetic acid and CH₂Cl₂ (v/v=1:5). The organic extract was washed with water (300 mL×3) and then evaporated to dryness to generate a blue solid. The residual high boiling point hexanol remaining in the solid was removed by repeatedly dispersing the solid in 40 mL of CH₂Cl₂/petroleum ether mixture (v/v=1:3) and decanting the brownish solvent. The blue crude product was passed through a silica gel pad (eluent: CH₂Cl₂/MeOH=100/1) to give as a blue solid, which was further recrystallized in a 100 mL MeOH/CH₂Cl₂ mixture (v/v=2:1) to produce dark blue crystals (1.97 g, yield 41%). ¹H NMR (600 MHz, 298K, deuterated chloroform (CDCl₃)) δ=10.56 (s, 4H), 9.91 (s, 4H), 9.81 (s, 4H), 9.36 (s, 4H), 9.07 (d, J=7.9 Hz, 8H), 8.41 (d, J=7.9 Hz, 8H), 7.40 (d, J=7.7 Hz, 7H), 6.61 (d, J=7.7 Hz, 8H), 3.31 (s, 12H), 1.98 (s, 12H). ¹³C NMR (150 MHz, 298 K, CDCl₃) δ=146.65, 144.01, 138.11, 136.39, 135.37, 133.20, 131.72, 129.63, 128.19, 127.35, 123.79, 122.46, 22.87, 21.51. HRMS (ESI): m/z calcd for C₃₈H₃₈N₈O₈S₈: 1924.2488; found: 1924.2495 [M+H]⁺.

(125) FIG. 9 illustrates ¹H NMR (600 M, 298 K, CDCl₃) spectrum of CoPc-NHTs and FIG. 10 illustrates ¹³C NMR (150 M, 298 K, CDCl₃) spectrum of CoPc-NHTs.

(126) Synthesis of CoOAPc. To a 50 mL round bottom flask charged with 1.0 g (0.51 mmol) of 2,3,9,10,16,17,23,24-octatosylaminophthalocyanine Co(II), a mixture of 3 mL of deionized water and 30 mL of concentrated H₂SO₄ was added. The mixture was then heated at 110° C. for about 1.5 hour. The reaction was then cooled to room temperature, and the reaction mixture was poured into ice-water (50 mL). The dark blue precipitate was collected by centrifugation and the solid was washed thoroughly with deionized water (20 mL×2), 10% NaOH (20 mL×3), deionized water (20 mL×3), and acetone (20 mL×3) sequentially with the help of a vortex. The product CoOAPc was obtained as a purple to black powder (325 mg, 91%). ¹H NMR (600 MHz, DMSO-d₆): δ=11.61 (br, 8H), 6.05 (s, 16H). HRMS (ESI): m/z calcd for C₃₂H₂₅CoN₁₆: 692.1779; found: 692.1772 [M+H]⁺.

(127) FIG. 11 illustrates ¹H NMR (600 M, 298 K, DMSO-d₆) spectrum of CoOAPc.

(128) Synthesis of CuOAPc. The synthetic scheme toward CuOAPc ligand is shown below in Scheme 3.

(129) ##STR00003##

(130) Synthesis of 2,3,9,10,16,17,23,24-octatosylaminophthalocyanine Cu(II) (CuPc-NHTs). The synthesis of the 2,3,9,10,16,17,23,24-octatosylaminophthalocyanine Cu(II) was adapted from the literature. To a suspension of 1.050 g (2 mmol) of N,N'-(4,5-dicyano-1,2-phenylene)bis(4-methylbenzenesulfonamide) and 0.122 g of Cu(OAc)₂ (0.674 mmol, 0.3 eq) in a glass pressure vessel, 5 mL of n-hexanol was added. The system was purged with N₂ for 10 minutes and then sealed. The reaction vessel was heated at 180° C. for 36 hours. The reaction was then allowed to cool to room temperature. Once at room temperature, the reaction mixture was dissolved in 66 mL of a mixed solvent system of acetic acid and CH₂Cl₂ (v/v=1:10). The organic extract was washed with water (50 mL×3) and then evaporated to dryness to provide an emerald-colored oil. The residual hexanol remaining in the oil was removed by repeatedly dispersing in 154 mL of a mixture CH₂Cl₂/petroleum ethers (v/v=1:10) and decanting the brownish solvent. The

solids were further purified by column chromatography (silica gel, eluent: CH₂Cl₂/MeOH=19/1). The solids were further purified by recrystallization in a CH₂Cl₂/MeOH mixture (v/v=3:1) to give a blue solid (35.3 mg, yield 3.26%). ¹H-NMR (500 MHz, CDCl₃) no proton peaks were observed in low-field NMR experiments. HRMS (ES⁺) calculated for C₈₈H₇₂CuN₁₆O₁₆S₈ 1928.24 [M^{sup.}+]; found 1930.24 m/z.

(131) FIG. 12 illustrates high resolution mass spectroscopy analysis of CuPc-NHTs. Electro-spray ionization was used in conjunction with TOF MS. The monomer ionized as M/1, and the mass of the parent peak was 1930.24 g/mol.

(132) Synthesis of CuOAPc. To a 50 mL round bottom flask charged with 47.4 mg of 2,3,9,10,16,17,23,24-octatosylaminophthalocyanine Cu(II), 142.2 μL of deionized water and 1.42 mL of concentrated H₂SO₄ were successively added. The mixture was then heated to 110° C. for 50 minutes under N₂. The reaction was then cooled to room temperature, and the reaction mixture was poured into ice-water (25 mL). The dark green precipitate was collected by centrifugation and the solid was washed thoroughly with deionized water (20 mL×3), 10% KOH (20 mL×3), deionized water (20 mL×3), and acetone (20 mL×3). The product was obtained as a dark purple to black powder (yield: 4.4 mg, 25.7%). ¹H-NMR (500 MHz, CDCl₃) no proton peaks were observed in low-field NMR experiments.

(133) Synthesis of MPc-Pyr-COFs. The synthetic scheme toward FePc-Pyr-COF is shown below in Scheme 4.

(134) ##STR00004##

(135) General synthesis of FePc-Pyr-COF. To prepare COFs, 7 mg (0.0077 mmol, 1 eq.) of FePcOAc and desired solvent (see Solvent System, Table 1) were added to a 15 mL glass pressure vessel. The vessel was purged for 10 min with nitrogen. After degassing, 200 μL of pTSA (3.5 M) was added to the reaction. Pyrenetetraone 4 mg (0.0154 mmol, 2 eq.) was then added under N₂. The reaction was allowed to degas with N₂ for an additional 5 min before being sealed and heated to 170° C. for 3 days. The reaction was allowed to cool to room temperature before being poured into acetone and centrifuged. The solids were washed with acetone (20 mL×3), 10% KOH (20 mL×3), deionized water (20 mL×3), and acetone (20 mL×3). The solids were dried overnight under a stream of nitrogen.

(136) TABLE-US-00001 TABLE 1 Conditions used to synthesize CoPc-Pyr-COF. Entry Solvent System [NiOAPc]/mM Catalyst Temperature/° C. Time 1 DCB/DMAC (1/1) 2 H₂SO₄ (9 eq) 180 7 d 2 NMP/mesitylene(2/1) 3 pTSA (s, 290 eq) 180 7 d 3 NMP/mesitylene(2/1) 3 H₂SO₄ (9 eq) 180 3 d 4 NMP/mesitylene(2/1) 12 3.5M pTSA (aq, 35 eq) 180 3 d 5 NMP/mesitylene(2/1) 12 3.5M pTSA (aq, 35 eq) 175 7 d 6 NMP/mesitylene(2/1) 12 3.5M pTSA (aq, 35 eq) 175 7 d 7 NMP/mesitylene(2/1) 12 H₂SO₄ (9 eq) 175 3 d 8 NMP/mesitylene(2/1) 12 6M HOAc (aq, 120) 120 6 d 9 NMP/mesitylene(2/1) 12 6M HOAc (aq, 120) 120 12 d 10 NMP 3.5 3.5M pTSA (aq, 35 eq) 175 7 d

(137) Synthesis of CoPc-Pyr-COF. The synthetic scheme toward CoPc-Pyr-COF is shown below in Scheme 5.

(138) ##STR00005##

(139) General synthesis of CoPc-Pyr-COF. 23 mg (0.02 mmol, 1 eq.) of CoPcOAc and desired solvent (see Solvent System, Table 1) to a 15 mL glass pressure vessel. The vessel was purged for 10 min with nitrogen. After degassing, acid catalyst was added to the reaction mixture. Pyrenetetraone 15 mg (0.04 mmol, 2 eq.) was then added under N₂. The reaction was allowed to degas with N₂ for an additional 5 min before being sealed and heated to 180° C. for 7 days. The reaction was allowed to cool to room temperature before being poured into acetone and centrifuged. The solids were washed with acetone (20 mL×3), 10% KOH (20 mL×3), deionized water (20 mL×3), and acetone (20 mL×3). The solids were dried overnight under a stream of nitrogen.

(140) Synthesis of CuPc-Pyr-COF. The synthetic scheme toward CuPc-Pyr-COF is shown below in Scheme 6.

(141) ##STR00006##

(142) General synthesis of CuPc-Pyr-COF. To prepare COFs, 7 mg (0.0077 mmol, 1 eq.) of CuPcOAc and desired solvent (see Solvent System, Table 1) to a 15 mL glass pressure vessel. The vessel was purged for 10 min with nitrogen. After degassing, acid catalyst was added to the reaction mixture. Pyrene-tetraone 4 mg (0.0154 mmol, 2 eq.) was then added under N.sub.2. The reaction was allowed to degas with N.sub.2 for an additional 5 min before being sealed and heated to 170° C. for 3 days. The reaction was allowed to cool to room temperature before being poured into acetone and centrifuged. The solids were washed with acetone (20 mL×3), 10% KOH (20 mL×3), deionized water (20 mL×3), and acetone (20 mL×3). The solids were dried overnight under a stream of nitrogen.

EXAMPLE 2.2

Powder X-Ray Diffraction (pXRD) Characterization of COFs

(143) Characterization of FePc-Pyr-COF by pXRD. FIG. 13 illustrates powder patterns obtained for various synthetic (Table 2) conditions leading to FePc-Pyr-COF. Powder patterns plotted against the computationally modeled eclipsed stacking structure. Peaks corresponding to petroleum jelly sued to secure the powder to the pXRD plate is labeled with an asterisk (*).

(144) TABLE-US-00002 TABLE 2 Conditions used to synthesize FePc-Pyr-COF. Condition Solvent System Catalyst 1 NMP 100% 3.5M pTSA (200 μL)

(145) Characterization of CoPc-Pyr-COF by pXRD. FIG. 14 illustrates experimental pXRD under various conditions (Table 1) and simulated pXRD for CoPc-Pyr-COF. Powder patterns plotted against the computationally modeled eclipsed stacking structure.

(146) Characterization of CuPc-Pyr-COF by pXRD. Powder patterns obtained for various synthetic conditions (Table 3) leading to CuPc-Pyr-COF. FIG. 15 illustrates powder patterns plotted against the computationally modeled eclipsed stacking structure.

(147) TABLE-US-00003 TABLE 3 Conditions used to synthesize CuPc-Pyr-COF. Condition Solvent System Catalyst 1 DMAC (7.5 mL): DCB (7.5 mL) H.sub.2SO.sub.4 2 DMAC (10 mL): DCB (5 mL) H.sub.2SO.sub.4 3 NMP (1.673 mL) PTSA (0.117 mL)

EXAMPLE 2.3

Scanning Electron Microscopy (SEM) of COF Materials

(148) FIG. 16 illustrates scanning electron microscopy of CoPc-Pyr-COF synthesized using condition 10 in Table 1. FIG. 17 illustrates scanning electron microscopy of CuPc-pyr-COF synthesized using condition 1 (Table 1). Working distance: 2.79 mm, SEM voltage: 20 kV-50 kV. FIG. 18 illustrates scanning electron microscopy of CuPc-pyr-COF synthesized using condition 2. Working distance: 2.79 mm, SEM voltage: 20 kV-50 kV.

EXAMPLE 2.4

Computational Study of Electronic Properties

(149) For computation of electronic properties of the COFs, including band structure, density of states, the functional generalized gradient approximation (GGA) with Perdew-Burke-Ernzerhof (PBE) was employed with an energy cutoff set at 530 eV. The SCF tolerance is set at ultra-fine level at 5.0×10^{-7} eV/atom. The Brillouin zones were sampled using a $2 \times 2 \times 4$ k-point mesh in the Monkhorst-Pack scheme.

(150) FIG. 19A illustrates calculated electronic band structure (left side) and density of state (right side) for FePc-Pyr-COF and FIG. 19B illustrates the corresponding first Brillouin zone and high-symmetry K-points. FIG. 20A illustrates calculated electronic band structure (left side) and density of state (right side) for CoPc-Pyr-COF and FIG. 20B illustrates the corresponding first Brillouin zone and high-symmetry K-points. FIG. 21A illustrates calculated electronic band structure (left side) and density of state (right side) for CuPc-Pyr-COF and FIG. 21B illustrates the corresponding first Brillouin zone and high-symmetry K-points.

(151) Without further elaboration, it is believed that one skilled in the art can, using the description herein, utilize the present disclosure to its fullest extent. The embodiments described herein are to be construed as illustrative and not as constraining the remainder of the disclosure in any way whatsoever. While the embodiments have been shown and described, many variations and modifications thereof can be made by one skilled in the art without departing from the spirit and teachings of the invention. Accordingly, the scope of protection is not limited by the description set out above, but is only limited by the claims, including all equivalents of the subject matter of the claims. The disclosures of all patents, patent applications and publications cited herein are hereby incorporated herein by reference, to the extent that they provide procedural or other details consistent with and supplementary to those set forth herein.

Claims

1. A method of sensing an analyte in a sample, said method comprising: exposing the sample to an electrode comprising a covalent-organic framework, wherein the covalent-organic framework is in the form of a square lattice comprising a plurality of metallophthalocyanine units that are linked to one another by aromatic linkers, wherein the aromatic linkers are conjugated; detecting a change in a property of the electrode; and correlating the change in the property to the presence or absence of the analyte.
2. The method of claim 1, wherein the metallophthalocyanine units are selected from the group consisting of octatosylaminophthalocyanine, octaamino-nickelphtalocyanine, nickelphtalocyanine, and combinations thereof.
3. The method of claim 1, wherein the metallophthalocyanine units comprise one or more metals selected from the group consisting of first row transition metals, divalent metals, transition metals, iron, nickel, copper, cobalt, zinc, manganese, platinum, palladium, gold, bismuth, chromium, magnesium, tin, and combinations thereof.
4. The method of claim 3, wherein the metallophthalocyanine units comprise nickel.
5. The method of claim 1, wherein the aromatic linkers comprise pyrenes.
6. The method of claim 5, wherein the pyrenes comprise pyrenetetraone.
7. The method of claim 1, wherein the aromatic linkers and the metallophthalocyanine units are linked through pyrazine rings.
8. The method of claim 7, wherein the pyrazine rings comprise phenazine.
9. The method of claim 1, wherein the covalent-organic framework has a fully aromatic and conjugated structure.
10. The method of claim 1, wherein the covalent-organic framework has a planar and two-dimensional structure.
11. The method of claim 1, wherein the analyte comprises a gaseous analyte.
12. The method of claim 11, wherein the gaseous analyte is selected from the group consisting of NH₃ (ammonia), H₂S (hydrogen sulfide), NO (nitric oxide), NO₂ (nitrogen dioxide), and combinations thereof.
13. The method of claim 1, wherein the detected change in the property of the electrode is a change in normalized conductance, and wherein the change in normalized conductance is characterized by the following formula: $-\Delta G/G_{sub.0} = -(I_{sub.0} - I)/I_{sub.0} \times 100\%$, where $-\Delta G/G_{sub.0}$ represents normalized conductance, $I_{sub.0}$ represents initial current, and I represents current at various points during measurement.
14. The method of claim 1, wherein the change in the property of the electrode is correlated to the presence or absence of the analyte by comparing the change in the property to the change in property of the electrode in response to known analytes.
15. The method of claim 1, wherein the covalent-organic framework exhibits a conductivity of at least 2.51×10^3 siemens per meter (S/m).

16. A covalent-organic framework comprising: a plurality of metallophthalocyanine units, that are linked to one another by aromatic linkers, wherein the aromatic linkers are conjugated, and wherein the covalent-organic framework is in the form of a square lattice.
 17. The covalent-organic framework of claim 16, wherein the metallophthalocyanine units are selected from the group consisting of octatosylaminophthalocyanine, octaamino-nickelphthalocyanine, nickelphthalocyanine, and combinations thereof.
 18. The covalent-organic framework of claim 16, wherein the metallophthalocyanine units comprise one or more metals selected from the group consisting of first row transition metals, divalent metals, transition metals, iron, nickel, copper, cobalt, zinc, manganese, platinum, palladium, gold, bismuth, chromium, magnesium, tin, and combinations thereof.
 19. The covalent-organic framework of claim 18, wherein the metallophthalocyanine units comprise nickel.
 20. The covalent-organic framework of claim 16, wherein the aromatic linkers comprise pyrenes.
 21. The covalent-organic framework of claim 20, wherein the pyrenes comprise pyrenetetraone.
 22. The covalent-organic framework of claim 16, wherein the aromatic linkers and the metallophthalocyanine units are linked through pyrazine rings.
 23. The covalent-organic framework of claim 22, wherein the pyrazine rings comprise phenazine.
 24. The covalent-organic framework of claim 16, wherein the covalent-organic framework has a fully aromatic and conjugated structure.
 25. The covalent-organic framework of claim 16, wherein the covalent-organic framework has a planar and two-dimensional structure.
 26. The method of claim 1, wherein the square lattice comprises a plurality of square apertures.
 27. The covalent-organic framework of claim 16, wherein the square lattice comprises a plurality of square apertures.
-